

# Ramadhan Adam Zome

Machine Learning Engineer · ML for Security · Representation Learning

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## EDUCATION

**Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI)** 2025 – 2027 (expected)  
*M.Sc. Artificial Intelligence and Machine Learning · Pan African University scholarship* Nairobi, Kenya

**Hiroshima University, Dependable Systems Laboratory** Aug 2026 – Jan 2027  
*Special Research Student: training the thesis's byte-level encoder on the lab's GPUs* Higashi-Hiroshima, Japan

**Kenyatta University** 2019 – 2023  
*B.Sc. Statistics and Programming · First Class Honours · county government scholarship* Nairobi, Kenya

## EXPERIENCE

**Graduate Researcher, Open-World Malware Detection (M.Sc. thesis)** 2025 – present  
*PAUSTI and Hiroshima University* Python, PyTorch, LIEF, HDBSCAN, AWS S3

- Built and pretrained a byte-level transformer that reads Windows executables by file region, with a learned structural attention bias; self-supervised on **9.9M malware binaries** (208K steps, 140 A100-hours).
- Engineered fault-tolerant ingestion of 9.9M files from S3 into 1,986 memory-mapped shards: process-isolated parsing, memory ceilings, atomic writes; peak **115K files/hour**, under 0.7% retried or skipped.
- Detected malware families never seen in training with **0.81 AUROC** (target 0.70), using multi-prototype families, regularized Mahalanobis distance and per-family thresholds.
- Showed with controlled baselines that pretraining adds signal: 0.80 family recall on 102 families, ahead of no-training references on 10 of 10 splits. Presenting the work in Hiroshima on 17 Oct 2026.

**Innovative Toll Solutions** Dec 2023 – Jul 2025  
*Data Analyst and Cybersecurity Associate* Nairobi, Kenya

- Cleaned, explored and modelled operational logs and transaction data; built dashboards and reporting pipelines used by operations teams.
- Ran log-level anomaly detection, alert triage and traffic-pattern analysis; supported pipeline resilience work after DDoS incidents.

## PROJECTS

**Production ML on Kubernetes** | FastAPI, k3d, GKE, Helm, GitHub Actions [github](#)

- Deployed a prediction API to GKE through GitHub Actions CI/CD, with a PostgreSQL StatefulSet for prediction logs, SOPS-encrypted secrets, Gateway API routing and Prometheus, Grafana and Loki monitoring.

**Explainable Deepfake Detection with RAG** | PyTorch, FAISS, FastAPI, React [live demo](#)

- Trained an Xception classifier on 140K real and GAN faces (about 97% held-out accuracy on 20K images) and added retrieval-augmented explanations, built without LangChain; FastAPI on Hugging Face Spaces, React on Vercel.

**Phish-Transformer** | PyTorch, Flask, Chrome MV3 [github](#)

- A 45K-parameter character-level transformer for phishing URLs (test AUC about 0.98) that runs on CPU behind a Flask API and warns users through a Chrome extension.

**Windows PE Internals: sco-pe and raw-pe** | C, NASM x86-64 [github](#)

- Wrote a PE parser in C with defensive bounds checks and unit tests (headers, sections, imports, exports, overlays), and hand-assembled a working 2,560-byte Win64 executable in NASM with no linker.

**Swahili GPT** | PyTorch, A100 [model](#)

- Trained a character-level GPT from scratch on 65M characters of Swahili news; used it to evaluate an A100 cloud platform for PAUSTI and wrote the decision report.

## SKILLS

**Languages** Python, C, SQL, Bash, x86-64 assembly (NASM), JavaScript

**ML** PyTorch, TensorFlow, scikit-learn, Hugging Face, FAISS, HDBSCAN, contrastive and self-supervised learning

**Systems** Docker, Kubernetes, Helm, GitHub Actions, GKE, FastAPI, PostgreSQL, Prometheus, Grafana, Linux

**Security** PE format, static and dynamic malware analysis, reverse engineering, REMnux, FLARE-VM

**Writing** Technical articles on PE internals, GPU memory and transformers: [medium.com/@ramadhanzome4](https://medium.com/@ramadhanzome4)